

CHAPTER 3 km reanalysis — GRIB2 archive contents

75 variables, 231 GRIB messages per hourly file. Generated from the converter's own registry (`wrf_era5_comparison.WRF_TO_ECMWF_PARAMID`), which is what the pipeline and its sanity check use, so this document cannot drift from what is actually written. Document generated on 2026-09-17. Every field here is one we can defend row by row. Where the WRF quantity is not exactly the ERA5 one, it is said so below rather than hidden; where it could not be made to correspond at all, the field was dropped instead of published under a name that would mislead (see §6).

1. What the files are

Model	WRF 4.1.1, domain d02, 49 vertical levels, model top 50 hPa
Horizontal grid	Mercator, 1641 × 1353 points, 3 km spacing (2 220 273 points/field)
Domain corners	23.472 N, 19.972 W → 60.001 N, 42.256 E; standard parallel 44.671 N
Earth shape	Sphere of radius 6 370 000 m (<code>shapeOfTheEarth=1</code>) — WRF's own sphere, declared explicitly
GRIB edition	GRIB2 , grid definition template 3.10 (Mercator), originating centre <code>ecmf</code>
Packing	<code>grid_ccsds</code> , 24 bits per value ; masked fields carry a proper GRIB bitmap
Parameter identity	ECMWF paramId / shortName throughout, so the archive is directly comparable with ERA5
File	one file per hour, ~770 MB (~19 GB/day, ~7 TB/year)
Naming	<code>ailam-an-cima-3km-{year}-{year}-1h-v1-{YYYYMMDD}{HH}.grib</code>
Period	2019, 2024 and 2025-01-01 → 2025-06-30, hourly (2020–2023 to follow)

2. Vertical levels

13 pressure levels, the same for every 3-D variable:

1000, 925, 850, 700, 600, 500, 400, 300, 250, 200, 150, 100, 50 hPa

They are the ERA5 set truncated at the model top: the 8 ERA5 levels above 50 hPa are above WRF's `P_TOP` and cannot be produced. Single-level fields sit on surface, `heightAboveGround` (2, 10, 100 and 200 m), `meanSea`, `entireAtmosphere` (column integrals) or, for the radiation budget at the top of the atmosphere, `nominalTop`.

3. Time convention — read this before indexing the archive

- **Instantaneous fields** use product definition template **0**: `dataDate/dataTime` are the valid time.
- **Accumulated fields** (`tp`, `tirf`, `sf`, `sro`, `ssro`, `ro`, and the whole radiation budget) use template **8**, as their ECMWF definitions require. They are accumulated ****from 00 UTC of the same day****: reference time 00 UTC, step 0–H. They are exactly zero in the 00 UTC file.

- **10fg** is the one exception: it is a **maximum over the preceding hour** (reference time H–1, step 0–1), and it is the maximum of the *resolved* 10 m wind, not a gust parameterisation.
- A file therefore mixes reference times on purpose. ****Index on validity (validityDate/validityTime), never on dataTime.****
- Accumulated messages carry generatingProcessIdentifier = 128 as the marker of the 00 UTC reference.

4. Declared approximations

Three fields are not exactly their ERA5 namesake, and are published with that said:

field	what ERA5 means	what we write
slor	slope of the sub-grid orography, a parameter of the form-drag scheme	the slope of the resolved 3 km terrain, $ \text{grad } z $
10fg	gust from a parameterisation (turbulent + convective)	the hourly maximum of the resolved 10 m wind
IEWS, INSS	the model's own stress components	the magnitude ρu^2 , projected on the 10 m wind direction

Two more deserve a note rather than a warning:

- **skt** is not a model output here (TSK was not written): it is inverted from the upward longwave, $\text{LWUPB} = \epsilon \sigma T^4 + (1-\epsilon) \text{LWDN}$, with the emissivity of the dominant land-use category. It is exact where the answer is known — over open water the model's skin temperature is the SST, and the inversion recovers it to **0.005 K** — and on snow-free land it matches the 0 cm soil level to better than 1 K RMSE. Over snow-covered points the emissivity the land-surface scheme actually used cannot be reconstructed, leaving about ± 0.8 K there.
- **sst** is constant within each 24 h run (SST_UPDATE=0), re-initialised daily.

5. Masked fields

Written with a GRIB bitmap rather than fake zeros, so missing means missing:

field	defined where
sst, ci	sea points only
tsn	cells whose snow cover exceeds 0.9 (≈ 29 mm water equivalent); out of season the field is legitimately empty
fal	daytime only (incoming shortwave above 50 W/m ²)

6. What is not in the archive

No TKE and no boundary-layer height (the PBL scheme is YSU, non-local); no ocean waves, currents, sea surface height or sea-ice thickness (no ocean or wave coupling); no convective precipitation (CU_PHYSICS=0, so it is identically zero and would be a dead field). Deliberately dropped rather than approximated:

- **soil moisture and soil temperature** (swvl1-4, stl1-4): the land-surface scheme carries point values at 0, 5, 20, 40, 160 and 300 cm, ERA5 averages over 0-7, 7-28, 28-100 and

100-289 cm. Writing the first four under the ERA5 names would put a 40 cm value under a name that means 1-2.9 m;

- **vegetation and soil type** (tv_l, tv_h, slt): the category numbers are WRF's MODIS-IGBP and STATSGO ones, while those paramIds carry ECMWF's own code tables — code 1 means "crops" to a reader of table 4.234 and "evergreen needleleaf forest" to us;
- **vegetation cover and leaf area index** (cv_l, cv_h, lai_{lv}, lai_{hv}): all rest on one dominant category per cell, while ERA5 lets low and high vegetation coexist, and the model's vegetation fraction is a seasonal *green* fraction, not a static cover fraction;
- **fractional land cover**: it lives in a static file that no longer exists.

The full reasoning, and the routes by which some of it could still be obtained, is in MISSING_VARIABLES.pdf.

7. The variables

Descriptions and units are the ECMWF ones. "Time" says how the field is sampled. There are 75 rows and 74 distinct shortNames, because z is written twice: as geopotential on the pressure levels and as the surface geopotential (orography). variables=75 messages=231

Pressure levels (13) — 13 variables, 169 messages

shortName	paramId	Description	Units	Level	Time
cc	248	Fraction of cloud cover	(0-1)	13 levels	instantaneous
ciwc	247	Specific cloud ice water content	kg/kg	13 levels	instantaneous
clwc	246	Specific cloud liquid water content	kg/kg	13 levels	instantaneous
crwc	75	Specific rain water content	kg/kg	13 levels	instantaneous
cswc	76	Specific snow water content	kg/kg	13 levels	instantaneous
q	133	Specific humidity	kg/kg	13 levels	instantaneous
r	157	Relative humidity	%	13 levels	instantaneous
t	130	Temperature	K	13 levels	instantaneous
u	131	U component of wind	m/s	13 levels	instantaneous
v	132	V component of wind	m/s	13 levels	instantaneous
w	135	Vertical velocity (pressure)	Pa/s	13 levels	instantaneous
wz	260238	Geometric vertical velocity	m/s	13 levels	instantaneous
z	129	Geopotential	m ² /s ²	13 levels	instantaneous

Height above ground — 11 variables, 11 messages

shortName	paramId	Description	Units	Level	Time
100u	228246	100m U component of wind	m/s	100 m	instantaneous
100v	228247	100m V component of wind	m/s	100 m	instantaneous
10fg	49	10m wind gust (hourly max of resolved wind)	m/s	10 m	max over the previous hour
10u	165	10m U component of wind	m/s	10 m	instantaneous
10v	166	10m V component of wind	m/s	10 m	instantaneous
200u	228239	200m U component of wind	m/s	200 m	instantaneous
200v	228240	200m V component of wind	m/s	200 m	instantaneous
2d	168	2m dewpoint temperature	K	2 m	instantaneous
2r	260242	2m relative humidity	%	2 m	instantaneous
2t	167	2m temperature	K	2 m	instantaneous
vwsh	260068	Vertical speed shear (bulk 10-100 m)	s ⁻¹	100 m	instantaneous

Mean sea level — 1 variables, 1 messages

shortName	paramId	Description	Units	Level	Time
msl	151	Mean sea level pressure	Pa	mean sea level	instantaneous

Total column — 10 variables, 10 messages

shortName	paramId	Description	Units	Level	Time
hcc	188	High cloud cover	(0-1)	column	instantaneous
lcc	186	Low cloud cover	(0-1)	column	instantaneous
mcc	187	Medium cloud cover	(0-1)	column	instantaneous
tcc	164	Total cloud cover	(0-1)	column	instantaneous

shortName	paramId	Description	Units	Level	Time
tciw	79	Total column cloud ice water	kg/m ²	column	instantaneous
tclw	78	Total column cloud liquid water	kg/m ²	column	instantaneous
tcrw	228089	Total column rain water	kg/m ²	column	instantaneous
tcsw	228090	Total column snow water	kg/m ²	column	instantaneous
tcw	136	Total column water	kg/m ²	column	instantaneous
tcwv	137	Total column water vapour	kg/m ²	column	instantaneous

Surface / single level — 33 variables, 33 messages

shortName	paramId	Description	Units	Level	Time
al	174	Albedo (climato-logical, snow-free background)	(0-1)	surface	instantaneous
ci	31	Sea ice area fraction	(0-1)	surface	instantaneous
fal	243	Forecast albedo (upward/downward SW at surface)	(0-1)	surface	instantaneous
fsr	244	Forecast surface roughness	m	surface	instantaneous
IEWS	229	Instantaneous eastward turbulent surface stress	N/m ²	surface	instantaneous
INSS	230	Instantaneous northward turbulent surface stress	N/m ²	surface	instantaneous
lsm	172	Land-sea mask	(0-1)	surface	instantaneous
ro	205	Runoff (surface + sub-surface)	m	surface	accumulated since 00 UTC
rsn	33	Snow density	kg/m ³	surface	instantaneous
sd	141	Snow depth (water equivalent)	m	surface	instantaneous
sdor	160	Standard deviation of orography	m	surface	instantaneous

shortName	paramId	Description	Units	Level	Time
sf	144	Snowfall (water equivalent)	m	surface	accumulated since 00 UTC
skt	235	Skin temperature (inverted from upward longwave)	K	surface	instantaneous
slor	163	Slope of orography (resolved, 3 km)	Numeric	surface	instantaneous
snowc	260038	Snow cover	%	surface	instantaneous
sp	134	Surface pressure	Pa	surface	instantaneous
src	198	Skin reservoir content	m	surface	instantaneous
sro	8	Surface runoff	m	surface	accumulated since 00 UTC
ssr	176	Surface net solar radiation	J/m ²	surface	accumulated since 00 UTC
ssrc	210	Surface net solar radiation, clear sky	J/m ²	surface	accumulated since 00 UTC
ssrd	169	Surface solar radiation downwards	J/m ²	surface	accumulated since 00 UTC
ssrdc	228129	Surface solar radiation downwards, clear sky	J/m ²	surface	accumulated since 00 UTC
ssro	9	Sub-surface runoff	m	surface	accumulated since 00 UTC
sst	34	Sea surface temperature	K	surface	instantaneous
str	177	Surface net thermal radiation	J/m ²	surface	accumulated since 00 UTC
strc	211	Surface net thermal radiation, clear sky	J/m ²	surface	accumulated since 00 UTC
strd	175	Surface thermal radiation downwards	J/m ²	surface	accumulated since 00 UTC
strdc	228130	Surface thermal radiation downwards, clear sky	J/m ²	surface	accumulated since 00 UTC

shortName	paramId	Description	Units	Level	Time
tirf	235015	Time integral of rain flux (rain only)	kg/m ²	surface	accumulated since 00 UTC
tp	228	Total precipitation	m	surface	accumulated since 00 UTC
tsn	238	Temperature of snow layer	K	surface	instantaneous
z	129	Geopotential (surface)	m ² /s ²	surface	instantaneous
zust	228003	Friction velocity	m/s	surface	instantaneous

Top of atmosphere / most unstable parcel — 7 variables, 7 messages

shortName	paramId	Description	Units	Level	Time
mucape	228235	Most-unstable CAPE	J/kg	most unstable parcel	instantaneous
mucin	228236	Most-unstable CIN	J/kg	most unstable parcel	instantaneous
tisr	212	TOA incident solar radiation	J/m ²	top of atmosphere	accumulated since 00 UTC
tsr	178	Top net solar radiation	J/m ²	top of atmosphere	accumulated since 00 UTC
tsrc	208	Top net solar radiation, clear sky	J/m ²	top of atmosphere	accumulated since 00 UTC
ttr	179	Top net thermal radiation	J/m ²	top of atmosphere	accumulated since 00 UTC
ttrc	209	Top net thermal radiation, clear sky	J/m ²	top of atmosphere	accumulated since 00 UTC